

Main tables

Joint treatment and source-discrepancy modeling with locally robust commensurate Bayesian additive regression trees

Table 1: Descriptive joint treatment and control comparison across five Gaussian conditions. Each entry gives RMSE / 95% interval coverage count / ATE convergence flags, with 40 datasets per condition. The flags count method–dataset groups with $\hat{R} > 1.05$ or bulk effective sample size below 400; they do not count individual chains. Comparative estimates remain provisional because of these diagnostics.

Condition	Joint LRC	Joint source-BART	Separated source-BART
Compatible	0.1750 / 38 / 0	0.2133 / 39 / 1	0.2064 / 37 / 1
One-variable	0.2068 / 38 / 3	0.2155 / 38 / 2	0.2105 / 37 / 0
Two-variable	0.2151 / 38 / 3	0.2147 / 38 / 2	0.2113 / 37 / 7
Heterogeneous	0.2092 / 38 / 8	0.2112 / 38 / 4	0.1917 / 39 / 5
Null	0.2298 / 36 / 4	0.2150 / 38 / 3	0.2100 / 38 / 8

Table 2: Characteristics of the harmonized multiple-myeloma analysis cohorts.

Characteristic	EloKRd	UCMM
Participants	30	200
Age (years), median (IQR)	62.8 (57.4, 70.2)	60.6 (54.0, 66.9)
Male	21 (70.0)	109 (54.5)
Race		
White	23 (76.7)	131 (65.5)
Black	4 (13.3)	58 (29.0)
Other	3 (10.0)	11 (5.5)
Hispanic or Latino	1 (3.3)	10 (5.0)
High-risk cytogenetics	15 (50.0)	51 (25.5)
ASCT	9 (30.0)	174 (87.0)
Progression or death	13 (43.3)	127 (63.5)
All-cause death	10 (33.3)	43 (21.5)

Entries are n (%) except where indicated. IQR, interquartile range; ASCT, autologous stem-cell transplantation. ASCT is a postinduction characteristic and is not presented as a pretreatment baseline variable. All analysis covariates were complete in both cohorts.

Table 3: Five-year restricted mean survival time (RMST) in the application.

Method	RMST ratio (95% interval)	RMST difference (95% interval)	EloKRd treatment RMST	EloKRd hypothetical control RMST	UCMM control RMST
<i>PFS</i>					
KM	1.040 (0.862, 1.254)	0.143 (-0.548, 0.834)	3.705	—	3.562
lrcBART	0.997 (0.799, 1.229)	-0.009 (-0.768, 0.714)	3.521	3.535	—
BART-CP	0.988 (0.796, 1.215)	-0.042 (-0.760, 0.690)	3.527	3.580	—
AFT-CP	0.901 (0.708, 1.105)	-0.338 (-1.080, 0.335)	3.069	3.408	—
hier. AFT	1.062 (0.789, 1.467)	0.192 (-0.834, 1.138)	3.355	3.148	—
<i>OS</i>					
KM	0.891 (0.776, 1.023)	-0.505 (-1.077, 0.068)	4.122	—	4.627
lrcBART	0.898 (0.765, 1.024)	-0.453 (-1.065, 0.101)	3.995	4.456	—
BART-CP	0.898 (0.769, 1.017)	-0.448 (-1.048, 0.073)	3.985	4.439	—
AFT-CP	0.809 (0.654, 0.951)	-0.814 (-1.521, -0.203)	3.449	4.268	—
hier. AFT	0.961 (0.761, 1.387)	-0.153 (-1.085, 1.073)	3.807	3.953	—

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RMST and differences are in years. For KM, EloKRd treatment RMST and UCMM control RMST are observed cohort estimates. For adjusted methods, EloKRd treatment and hypothetical control RMSTs are model-based estimates standardized to EloKRd covariates; UCMM control RMST is not reported. A dash denotes an inapplicable estimate. KM is an unadjusted comparison with confidence intervals; model-based intervals are equal-tailed credible intervals. Arm summaries are posterior medians, except KM estimates. The median of a ratio or difference need not equal the ratio or difference of marginal medians. Primary lrcBART uses $H_f = 10$, $H_g = 5$, $w = 1$, requested ESS 30. The hier. AFT comparator uses hierarchical discrepancy-variance prior scale $s_\tau^2 = 0.05$, where $\tau_\alpha^2, \tau_\beta^2 \sim \text{IG}(3/2, 3s_\tau^2/2)$.

Table 4: UCMM-informed prior ESS for the 30 EloKRd patient covariate profiles, with UCMM covariate-combination counts (PFS).

Age group	Age	Sex	Race	Hispanic/ Latino	Cytogenetic risk	ASCT	UCMM cohort n	Prior ESS
< 50	42.8	Male	White	No	High	Yes	0	11.0
< 50	45.4	Male	White	No	High	No	0	6.0
< 50	48.4	Male	Black	No	Standard	No	0	6.7
50–59	50.7	Female	White	No	High	No	0	4.7
50–59	51.7	Male	Other	No	Standard	Yes	1	7.1
50–59	55.7	Male	White	No	Standard	No	2	8.3
50–59	56.6	Male	White	No	Standard	Yes	22	21.2
50–59	57.2	Male	White	No	Standard	Yes	22	21.2
50–59	58.1	Female	White	No	Standard	Yes	17	15.8
50–59	59.0	Female	Black	No	High	No	0	5.1
50–59	59.4	Male	White	No	Standard	Yes	22	21.7
50–59	59.5	Male	White	No	High	Yes	6	13.1
50–59	59.7	Female	White	No	Standard	No	0	7.0
60–69	61.6	Female	White	No	High	No	0	5.5
60–69	62.4	Male	White	No	High	No	2	7.2
60–69	63.2	Male	Other	No	High	No	0	4.3
60–69	65.6	Male	White	No	Standard	Yes	18	24.5
60–69	67.4	Male	White	No	High	No	2	7.2
60–69	67.7	Female	White	No	Standard	No	0	7.6
60–69	67.9	Male	Other	Yes	Standard	No	0	3.8
60–69	69.7	Male	White	No	High	No	2	7.2
≥ 70	70.0	Male	White	No	Standard	No	4	9.4
≥ 70	70.3	Male	White	No	High	No	0	7.1
≥ 70	70.4	Male	White	No	High	Yes	2	13.5
≥ 70	70.4	Male	White	No	High	No	0	7.1
≥ 70	70.9	Male	White	No	High	No	0	7.1
≥ 70	72.1	Male	White	No	Standard	No	4	9.5
≥ 70	76.1	Female	White	No	High	No	1	5.6
≥ 70	77.0	Female	Black	No	Standard	No	3	7.2
≥ 70	81.2	Female	Black	No	Standard	No	3	7.0

Each row is one EloKRd patient covariate profile. Age groups are < 50, 50–59, 60–69 and ≥ 70 years. Age is rounded to one decimal year for display; calculations use exact age. UCMM cohort n is the number of patients among all 200 triplet-treated UCMM controls matching that row’s age group, sex, race, Hispanic/Latino status, cytogenetic risk and ASCT. Counts repeat when EloKRd rows share the same displayed combination and should not be summed down the table. Prior ESS is informed by the full 200-patient triplet UCMM cohort and is expressed in uncensored normal-reference units for the conditional mean log-PFS at the same profile. The same residual variance and globally selected PFS scale are used for all profiles. Values do not measure outcome compatibility and do not sum or average to overall ESS. ASCT is postinduction.